Tasks

2.28.1 I. Use numpy methods only, not standard Python loops, generators, math functions, etc.

2.28.2 II. Task numbers correspond to your group list.

2.29 Creating 1D and 2D Arrays

1. Create a vector with `array`.
2. Create a vector with `arange`.
3. Create a vector with `linspace`.
4. Create a vector with `zeros`.
5. Create a vector with `full`.

2.30 The `random` Module

6. Create a 2D array of random integers with `randint`.
7. Create a 2D array with `uniform`.
8. Create a 2D array with `normal`.

2.31 Array Attributes and Operations

9. Display `dtype`, `ndim`, `shape`, `size`, `itemsize`, `nbytes`.
10. Select and modify elements using slicing.
11. Perform arithmetic operations on arrays.

2.32 Distance, Norm, Dot Product

12. Compute ρ_1 (Manhattan) distance.
13. Compute ρ_2 (Euclidean) distance.
14. Compute ρ_∞ (Chebyshev) distance.
15. Compute $\|a\|_1$ norm.
16. Compute $\|a\|_2$ norm.
17. Compute $\|a\|_\infty$ norm.
18. Compute dot product.

2.33 Matrix Operations

19. Arithmetic and matrix product (`np.dot`).
20. Determinant.
21. Eigenvalues.
22. Eigenvalues and eigenvectors.
23. Inverse.
24. Norm.

2.34 Aggregation

25. Sum, mean, min.
26. Product, std, argmin.
27. Mean, variance, median.

2.35 Comparisons, Masks, Boolean Logic

28. Count elements in [-5, 5].
29. Check if any elements > 100.
30. Check if all elements are positive.
31. Select elements in [0, 10].
32. Replace negative elements with 0.

2.36 Adding a Column

33. Add column of column means.
34. Add column of squared values from first column.
35. Add column of row maxima.

2.37 Sorting

36. Sort rows.
37. Sort columns.
38. Sort all elements (`axis=None`).
39. Sort one column only.
40. Sort one row in reverse order.

2.38 Saving and Loading

41. Binary: `np.save` / `np.load`
42. Text: `np.savetxt` / `np.loadtxt`